



Recognising special number sequences

1 The next term in the sequence 8, 2, 10, 12, ... is found by adding the two previous terms. The next term is 22. Fill in the blank

2 Which of these could be an arithmetic sequence? Tick **2** correct answers

☒ 76, 99, 122, 145, ...

☐ -14, -6, 6, 14, ...

☐ 12.4, 16.2, 20.1, 23.8, ...

☒ $\frac{4}{3}, \frac{3}{2}, \frac{5}{3}, \frac{11}{6}, \dots$

☐ $\frac{1}{8}, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \dots$

3 Which of these could be a geometric sequence? Tick **3** correct answers

☐ -108, 36, 12, -4, ...

☒ 4, -8, 16, -32, ...

☒ 1000, 200, 40, 8, ...

☒ 27, 36, 48, 64, ...

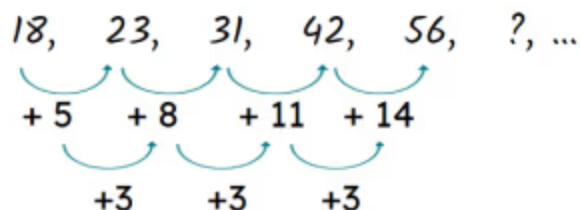
☐ 10, 20, 30, 40, ...

- 4 Match the first four terms of each sequence with the rule which could describe it. Write the correct letter in each box

a	$a, 2a, 4a, 8a, \dots$
b	$a, 2a, 4a, 7a, \dots$
c	$2a, 4a, 6a, 8a, \dots$
d	$3a, 3a^2, 3a^3, 3a^4, \dots$
e	$a + 1, 2a + 2, 3a + 3, 4a + 4, \dots$

d	geometric sequence with common ratio a
b	sequence which starts by adding a with second difference $+ a$
a	geometric sequence with common ratio 2
e	linear sequence with common difference $a + 1$
c	linear sequence with common difference $2a$

- 5 This sequence starts by adding 5 and has a common second difference of 3. The next term in the sequence is 73. Fill in the blank



- 6 Which of these could be the first four terms of a sequence with a common second difference? (They are called quadratic sequences). Tick 3 correct answers

- ☐ 5, 8, 11, 14, ...
- ☒ 12, 14, 18, 24, ...
- ☐ 9, 11, 15, 23, ...
- ☒ 4, 7, 11, 16, ...
- ☒ 8, 9, 13, 20, ...

